L2. Scientific Inference

Inference

What is it?

The process of reaching a conclusion from some facts or evidence.

Premise

What is it?

Premise is the starting assumption we take as true for the given problem. it's essentially a statement.

Example:

All balls are green

This object weighs 100kg

The moon is round

Different types of inferences

Direct inference

- Moves from a statement about a sample to a statement about a population.

Examples:

- **Premise**: The sky is blue **and** the grass is green → **Inference**: Therefore, the sky is blue.
- Seeing that

Generalization

What is it?

> We see several cases of something, and we conclude a general rule.

Form:

- We have seen: A, B, C are like this
- So we think: all (or most) things of this kind are like this

Example:

- You see 10 swans, and all of them are white.
- You conclude:
  "All swans in the world are white."

This is not guaranteed to be true (maybe a black swan exists), but it feels reasonable based on what weve seen.

Projection

We have seen a pattern so far, and we **project** it onto a **new case** (often in the future or somewhere else).

Form:

- So far, all Xs we've seen have property Y
- There is a new X we haven't checked yet
- We guess: the new X also has property Y

Example 1 (future):

- Every morning so far, the sun has risen in the east.
- Tomorrow is another morning.
- You conclude:
  "Tomorrow the sun will rise in the east."

Example 2 (medicine):

- A medicine helped many patients with the same illness.
- A new patient has that illness.
- You conclude:
  "This medicine will probably help this new patient too."

Projection is about applying a pattern to a new, unobserved case.

Modus Ponens (deductive)

“If P then Q. P is true. So Q must be true.”

This is a **valid deductive rule**: if the premises are true (i.e premises=statements), the conclusion must be true.

Form:

1. If P, then Q
2. P
3. Therefore, Q

Example:

1. If it is raining, then the streets are wet.
2. It is raining.
3. Therefore, the streets are wet.

If (1) and (2) are true, (3) cannot be false.

Modus tollens(deductive)

If P then Q. Q is not true. So P cannot be true. Like Modus pollens it is also deductive. But here we are doing the inversion. If P→ Q and Q is false then P cannot be true

Form:

1. If P, then Q
2. Not Q
3. Therefore, not P

Example:

1. If the fire alarm is working, then it will sound when there is smoke.
2. There is smoke, but the alarm is not sounding.
3. Therefore, the fire alarm is not working.

Again, if (1) and (2) are true, the conclusion (3) must be true.

Deductive Inferences

What it is:

Deductive inference is a type of reasoning where we start with **general statements or rules**, and from them we derive a conclusion that must be true **if the premises are true**. Deduction does **not add new information**, it simply *unpacks* what is already contained in the premises. Often times its pretty clear from the statement what will follow, its logical.

Example:

1. All humans are mortal.
2. Socrates is a human.
3. Therefore, Socrates is mortal.

Another example:

- All stars burn

- Our sun is a star
- Therefore our sun burns.

There is no way for the conclusion to be false if both premises(statements) are true. That is why deductive reasoning is described as truth-preserving.

The following are Deductive inference:

- Modus Ponens
- Modus Tollens

Inductive Inference

Inductive inference is a type of reasoning where we move from **specific observations** to a **general conclusion**. It goes beyond the information we already have, which means it can be **useful but never guaranteed to be true**. Induction is the kind of reasoning we use in everyday life, science, and prediction. Unlike deductive Inference the conclusion is *probable(might be false)*, not logically certain.

Example:

- You observe that every cat you have seen so far has whiskers.
- You conclude: "All cats have whiskers."

This may be true, but it is not logically guaranteed. One day you might see a cat without whiskers (e.g., due to injury or mutation), which would prove the generalization false. Induction is useful because we cannot observe everything, but we can still make reasonable expectations about the world.

Another example (science):

- Many experiments show that metals expand when heated.
- Scientists conclude: "All metals expand when heated."

  This is not absolutely certain, but strong evidence makes it reasonable to believe.

David Humes critique of inductive inference

Hume asks whether we can justify using induction - that is, whether we have a good reason to trust it. His answer is no, and he gives the following reasoning:

1. We want to justify inductive reasoning

Induction is something we use all the time, for example:

- "The sun has risen every day so far, so it will rise tomorrow."
- "Bread has always fed me, so it will feed me next time I eat it."

1. We cannot justify induction using deduction

   A deductive argument gives a conclusion that must be true if the premises are true.
   But inductive conclusions are not guaranteed - they go beyond what we have observed.
   So we cannot say:
   "The sun has risen every day so far, therefore it must rise tomorrow."
   Because logically, it might not. So deduction cannot save induction.

2. We also cannot justify induction by using more induction

   We might try to defend induction like this:
   "Induction has worked many times in the past, so it will work in the future."

   But this is itself an inductive argument - it assumes the future will be like the past.
   So we would just be using induction to justify induction, which is circular reasoning.

3. Therefore, induction cannot be justified

   Any attempt to defend induction either:

   - fails logically (deduction doesn't work), or
   - assumes the very thing it is trying to prove (using induction to justify induction).

   So Hume concludes we have no rational proof that induction is reliable, even though we all use it.

Example:

Inductive reasoning:

- Every time I have touched fire, it has burned me.
- So I believe fire will burn me next time too.

Hume's point:

- You cannot deductively prove that the fire will burn you next time - it's not logically guaranteed.
- You also cannot say "It always worked before, so it will work again," because that is already assuming that the future will be like the past - which is exactly what needs to be proven.

So the reasoning is useful, but not logically justified.

Foundationalism

Foundationalism is an argument against David Hume.

What is it?

Foundationalism is the view that knowledge must be built on **basic beliefs** that are certain, secure, and do not need further justification. These basic beliefs act like the foundation of a building: once they are in place, other beliefs can be justified by resting on them. This is used as an argument against David Hume, Foundationalists disagree with Hume because they

think there can be secure basic beliefs that do not need justification from past experience.

Example:

Think of the axioms in mathematics, like 0 is a natural number. Or the shortest distance between two points is a straight line. From this we can build theories.

Coherentism

What is it?

They believe that beliefs are not justified by a single foundation, like im happy because I saw my friend. Instead they believe that several beliefs working together and support each other make the belief possible.

Instead of:

> “I am happy because I saw my friend” (just one reason)
>
> Coherentism says:
>
> “I am happy because I saw my friend, I’ve been feeling good lately, the weather is nice, and I’m not stressed.”
>
> Several beliefs together make the main belief more justified

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Example:

Hume says we cannot prove the sun will rise tomorrow, because that is just assuming "the future will be like the past."

A coherentist replies:

We don't believe the sun will rise tomorrow because of one single rule ("it rose before, so it will rise again").
We believe it because this fits with many other beliefs that support each other, such as:

- "The laws of physics"
- "The movement of the Earth"
- "Observations from astronomy"
- "Past experience"
- "Scientific models that successfully predict sunrise times"

Even if one belief failed (e.g., an old astronomy model), the others still hold, so the belief remains justified inside the network.

So Coherentism does not try to escape Hume by proving induction, but instead says Hume's demand for a single foundation is the wrong way to think about justification.

The Hypothetical deductive method (HDM)

1. Formulate a hypothesis H
2. Deduce what observable consequences (C_i) from H
3. Test whether (C_i) is true or not.
4. If (C_i) is false infer that H is false.
5. If (C_i) is true, increase confidence in $`H`$.

Why is HDM both deductive and inductive?

Deductive:

If the hypothesis is true → then we should observe X .
We did not observe X therefore the hypothesis is false.
This is modus tollens, a valid deductive rule.

So deduction is used for falsifying hypotheses.

Inductive:

If the predicted observation does happen, we cannot logically prove the hypothesis is true - we only gain support for it. That is inductive reasoning:

The prediction came true → so the hypothesis is probably right.
But it could still be false in the future, so the support is not certain.

So induction is used for confirming hypotheses.

What is a good hypothesis?

- A statement that is either true or false.
  - Example: All metals expand when heated.

- A statement that is not necessarily true or false.
  - Example : Dark matter interacts only through gravity.

- Has generality or refers to the unobservable
  - Example: Electrons have a negative charge.
  - Example: All dogs need food to survive

What is a good observable sequence $`{(C_i)}`$?

- (C_i) must be observable directly or with the help of accurate measurements (e.g. microscope, X-ray, etc.) Something we can check essentially.

- Deduction must be valid. In other words we should be able to say H is false using {C_i}

- $ `{(C_i)}` $must be relevant for H.
    - **Example**:
      A hypothesis H can be that: **Plants need sunlight to survive**, the predicted consequence {C_i} can be : **If the hypothesis is true, then a plant kept in the dark will not grow well.**

Poppers Falsification

What is it?

Karl Popper argued that scientific theories can never be proven true, only proven false. A good scientific theory must make bold, testable predictions - predictions that could turn out to be wrong.

- If a prediction fails, the theory is falsified and must be rejected or revised.
- If a prediction succeeds, the theory is not proven true - it has only survived a test.

So for Popper, science progresses by eliminating false theories, not by confirming true ones.

Key idea:

Science should aim at falsifiability, not verification. A theory is scientific only if it can, in principle, be shown false.

Popper's slogan:
"One counterexample is enough."

Example:

- "All swans are white" is falsifiable - one black swan disproves it.
- "All stars are squared, our sun is round so it disproves it.

Argument against Poppers Falsification

Duhem-Quine Thesis

What is it?

> The idea is that when we test a hypothesis we never a test it alone. Instead we always rely on other assumptions, such as background theories, measurement tools, math, and auxiliary hypotheses (s maller hypothesis = auxiliary hypothesis).

So if a prediction fails, we cannot know for sure which part is wrong:

- the main hypothesis,
- one of the supporting assumptions,
- the experimental setup,
- the data interpretation, etc.

This means that we can’t simply say the hypothesis failed because the experiment failed. Instead we need to look into it more carefully and see if we made the wrong assumptions, wrong setup etc.

Example:

Suppose we test the hypothesis:
H: "This metal expands when heated."

We heat it and nothing happens. Does that mean H is false?

Not necessarily. Maybe:

- the thermometer was broken,
- the heating device malfunctioned,
- the sample wasn't actually metal,
- the temperature didn't rise enough,
- we used the wrong formula for expansion.

So the failed prediction does not automatically falsify the main hypothesis - something else in the background assumptions might be at fault.

Ad hoc modification

What is it?

Ad hoc modification is when a scientist changes or adds new assumptions to a theory only to protect it from being falsified, instead of improving the theory in a real scientific way.

Example:

Theory:
"All planets move in perfect circles."

Observation:
Mars does not move in a perfect circle.

Ad hoc fix:
"Mars is pushed by an invisible force that we cannot detect."

This saves the theory, but it does not explain anything new, predict anything useful, or make the theory stronger - it only protects it from being proven

wrong.

Hypothesis confirmation

Relevance

What is it?

When we test a hypothesis, **not all observations count as good evidence**. A piece of evidence is *relevant* only if it actually makes a difference to whether the hypothesis is likely true or false

Example:

Hypothesis: "This medicine reduces fever."

- Relevant evidence: Patients who take the medicine have lower temperature than those who don't.
- Irrelevant evidence: The pills are blue, the doctor likes the medicine, the patients slept well.

So relevance means: the evidence must be connected to what the hypothesis claims.

Underdetermination

What is it?

Underdetermination means that the same evidence can fit several different hypothesis, so the data by itself cannot tell us which explanation is the right one.

Example:

Data: The population of a city increased every year for 10 years.
Possible hypotheses:

1. "The birth rate is rising."
2. "More people are moving into the city."
3. "Death rates have fallen due to better healthcare."
4. "A combination of all three."

The same data fits multiple explanations - so the hypothesis is underdetermined by the evidence.

Observation and Measurement

Empiricism and Types of Observation

Empiricism

- **Empiricism**: the view that **sensory experience** is the ultimate source of knowledge.
- In the end, scientific claims must connect back to **what we can observe** with our senses, sometimes helped by instruments.

Direct Observation

- You use **only your own senses** to experience the event or object.
- **Examples:**
  - Seeing that a car is moving and not standing still.
  - Feeling that one bag is heavier than another.
  - Hearing that one sound is louder than another.
- **Limitations of direct observation**:

- It can be **mistaken**:
  - A color blind person can misjudge colors.
  - Strange lighting can change how colors or shapes appear.
- It is usually **qualitative**, not numerical:
  - You can tell "this is warmer than that", but not exactly *how much* warmer.
  - You can tell "this bag feels heavier", but not its exact weight in kilograms.

Aided Direct Observation

- You still perceive the **thing itself**, but with help from an instrument.
- **Examples**:
  - Glasses or contact lenses to see more sharply.
  - A microscope to see cells or bacteria.
  - A telescope to see distant planets or moons.
- These observations often depend on **background theory**:
  - To trust a telescope, we rely on a theory of **optics** that tells us how lenses bend light.
  - In Galileo's time, some scholars thought telescopes produced illusions, because they did not yet trust the theory.
- **Key idea:**
  - Aided direct observation is still "looking at the object", but the reliability of what you see depends on a theory of how the instrument works.

Indirect Observation

- You do **not** observe the phenomenon itself.

  Instead, you observe an **effect** that the phenomenon causes.
- General structure:

> Phenomenon (hidden or hard to observe)
>
> → causes some effect

> → we observe the effect and infer the phenomenon

- Examples:
  - We do not see **gravity**, but we see planets move in orbits and objects fall.
  - We do not see **elementary particles**, but we see **tracks** in a bubble chamber.
  - We do not see **temperature itself**, but we see mercury rise in a thermometer.
- **Indirect observations are always theory dependent**:
  - We need a theory that explains how the phenomenon produces the observable effect.
  - Without that theory, the effect would just be "bubbles in liquid" or "lines on a screen" and would not count as evidence for particles or fields.

Theory Dependence of Observation

- An observation is **theory dependent** if its **interpretation** depends on some accepted theory.
- Example with a microscope:
  - A scientist says "I see bacteria".
  - Strictly speaking, they see tiny shapes or dots in a microscope image.
  - It is the **theory of optics** plus **cell theory** and **staining techniques** that justifies calling those dots "bacteria".
- This shows:
  - Many observations, especially aided and indirect ones, are **not neutral**.
  - They already presuppose some theoretical assumptions in order to count as observations *of* something specific.

Logical Empiricism and Circularity

Logical Empiricism

- **Logical empiricism** is a philosophical movement with roughly these ideas:

    - Meaningful statements are either:
        - **empirically testable**, or
        - **logically or mathematically true**.
    - Metaphysical statements that cannot be tested are seen as **meaningless** for science.
    - Philosophy should clarify scientific language and logic, not introduce new untestable claims.
- Examples of how a logical empiricist classify statements:
    - "Electrons have negative charge" → meaningful, because it is testable in principle.
    - "God exists beyond space and time" → not meaningful for science, because it is not testable.
    - "2 + 2 = 4" → meaningful, because it is analytically or logically true.
- Problem highlighted in the chapter:
    - Logical empiricism needs a **clear distinction** between:
        - observational sentences, and
        - theoretical sentences.
    - If many observations are **theory dependent**, this clear separation becomes difficult or impossible.

Threat of Circularity in Indirect Observation

- In indirect observation, we:
    1. Use a **theory** to interpret what we see.
    2. Then use that interpreted observation as **support** for the same theory.
- This risks a kind of **circularity**.

Example from particle physics:

- We see **bubble tracks** in a chamber.
- Theory says:
    - particles cause ionization in the liquid,

    - ionization causes bubbles,
    - the shape of the track is related to charge, mass, and momentum of the particle.
- So we say "We observed a particle with these properties".
- But how do we know the theory about ionization and track shapes is correct?
    - Because it fits the bubble observations.

So:

- The **theory** is used to interpret the observation.
- Then the **interpreted observation** is used to support the theory.
- This is the **threat of circularity** in indirect observation.

The main lesson:

Observation and theory support each other, rather than having a simple one way order where observation comes first and theory comes later.

Operationalization and Operationalism

Operationalization

- **Operationalization** is the process of turning an **abstract or unobservable property** into something we can **measure**.
- Examples of properties of interest:
    - Temperature
    - Stress
    - Intelligence
    - Political attitude
    - Acidity
- Examples of operationalizations:
    - Temperature → height of mercury in a thermometer.
    - Stress → self report questionnaire and cortisol level in saliva.
    - Intelligence → score on an IQ test.